SABR Simulation for Risk Management
(Working Paper)

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1 Introduction

The SABR (Acronym for Stochastic alpha-beta-rho) model is a stochastic volatility model first studied by Hagan *et al*, 2002[6]. Due to its ability to replicate certain observed market phenomena, such as implied volatility smiles, frowns and skews, it has gained wide interest for the modeling of options. With its modeling power that can cover underlying assets that display normal, log-normal or CEV (Constant Elasticity of Variance) behavior, SABR is currently in extensive use in the area of fixed income derivatives.

Formally, assuming a filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, SABR is given by the following equations

$$dF_t = F_t^{\beta}\sigma_t dW_t \qquad \beta \in [0,1],\ F_0 > 0 \tag{1.1a}$$
$$d\sigma_t = \nu\sigma_t dZ_t \qquad \nu \geq 0,\ \sigma_0 > 0 \tag{1.1b}$$
$$[dZ_t, dW_t] = \rho dt \qquad \rho \in [-1,1] \tag{1.1c}$$

where $F_t, \sigma_t \in \mathcal{F}_t$, $[.,.]$ is the quadratic variation operator and dW_t, dZ_t are standard Wiener increments. Throughout the document, the term SABR volatility will refer to σ_t of Equation (1.1)

Pricing derivatives based on the SABR model is done using either a PDE (Partial Differential Equation) solver based on the two-dimensional discretization of the Kolmogorov backward equation, or a Monte Carlo simulation approach. PDE solvers tend to have much faster convergence. On the other hand, each run is effective in pricing a single option. Conversely, a set of Monte Carlo paths, once generated, can be used to price any number of derivatives as long as they share the same underlying asset and expiry date. This makes Monte Carlo simulation a more favorable approach in risk management, where risk measures associated with very large fixed income derivative portfolios need to be evaluated.

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2 Difficulties in SABR Modeling

To properly simulate the SABR model, we need to counter two essential difficulties. To explain the first difficulty, we use the following discretized version of (1.1)

$$\begin{aligned} F_t - F_{t-\Delta t} =& \bar{F}_t^{\beta} \sigma_? W_{\Delta t} \\ \sigma_t - \sigma_{t-\Delta t} =& \nu \bar{\sigma}_t Z_{\Delta t} \\ [W_{\Delta t}, Z_{\Delta t}] = \rho \Delta t & \end{aligned} \tag{2.1}$$

where $\bar{F}_t$ is an average value of F_s between time $t-\Delta t$ and t, $\bar{\sigma}_t$ is an average value of σ_s between time $t - \Delta t$ and t, and $\sigma_?$ is a representative value of σ between $t - \Delta t$ and t. The difficulty in choosing an appropriate value $\sigma_?$, when $\nu > 0$, stems from the fact that the continuous change of the value of σ is accompanied by a continuous change in F dependent on σ. Islah, 2009[9], showed that when $\beta \in \{0, 1\}$ or when $\rho = 0$, $\sigma_?$ can be exactly represented by the conditional average variance given as

$$\sigma_0 \sqrt{\Delta t \; I_{t-\Delta t}^{t}(\sigma_t)} \tag{2.2}$$

where

$$I_{t-\Delta t}^{t}(\sigma_t) := \frac{1}{\sigma_0^2 \, \Delta t} \int\limits_{t-\Delta t}^{t} \sigma_s^2 \, ds \, \Big|_{\sigma_t} \tag{2.3}$$

Choi *et al*, 2025[2], give the following equivalent functional

$$I_{t-\Delta t}^{t}(\hat{Z}_1) \sim \int\limits_0^1 \exp(2\nu \Delta t \, \hat{Z}_s) \, ds \, \big|_{\hat{Z}_1} \tag{2.4}$$

where $\hat{Z}_s \sim \mathcal{N}\left(-s\,\nu\sqrt{\Delta t}, \sqrt{s}\right)$. Although $I_{t-\Delta t}^{t}(\sigma_t)$ has a closed form PDF, as shown by Cai *et al*, 2017[4], there is no closed form for the CDF and its numerical integration is very difficult and highly error-prone. In their paper, Cai *et al* suggested the use of Laplace transform inversion to obtain an accurate and efficient representation of the CDF. We will be using a modified version of their Laplace inversion algorithm for the purposes of this paper.

The second difficulty in simulating a SABR process, is that the proxy for $\sigma_?$ given by (2.2) is no longer valid. No other exact proxy is currently available. As a result, one has to resort to plausible approximations such as those presented in [4],[2] and[6].

3 Sampling σ_t

In the process of simulating the SABR process, one may need to directly sample the volatility process itself. As per Equation (1.1b), σ_t follows a log-normal process. As a result, it can be sampled according to

$$\begin{aligned}\sigma_t =&\sigma_0 \, \exp(\nu \sqrt{t} \, \hat{Z}_t) \\ \hat{Z}_t \sim&\mathcal{N}\left(-\tfrac{1}{2}\nu \sqrt{t}, \, 1\right) \, .\end{aligned} \tag{3.1}$$

Thus, σ_t is sampled *via* a single Gaussian variate.

4 Sampling the Average Variance

As explained in Section 2, the average volatility variance $I_0^t(\sigma_t)$ is needed for the simulation of the price process. As discussed earlier, the sampling of $I_0^t(\sigma_t)$ is fraught with difficulty, due to the lack of a closed form representation of its CDF. For the sake of presentational simplicity, we may notationally drop the σ_t dependency of I_0^h, in the remainder of the text, without changing the fact.

In[2], Choi *et al* proposed an approach for the sampling of I_0^t by matching its CDF to that of shifted log-normal ($\mathcal{SLN}(\mu, \sigma^2, \lambda)$) process with matching first four moments. The $\mathcal{SLN}$ process is defined as

$$\begin{aligned}Y \sim&\mathcal{SLN}(\mu, \sigma^2, \lambda) := \\ Y \sim&\mu \left[(1-\lambda) + \lambda \exp(\sigma X - \frac{\sigma^2}{2})\right] \qquad \mu > 0, \sigma > 0, 0 < \lambda \leq 1 \, ,\end{aligned} \tag{4.1}$$

where X is a standard Normal variate, where μ and σ^2 are the mean and variance of the LN process and λ is the shift parameter. In their work, Choi *et al* suggested setting $\lambda = \frac{5}{6}$ and sampling I_0^t according to

$$\begin{aligned}I_0^t \sim \mathcal{SLN}&\left(\mu, \sigma^2, \frac{5}{6}\right) \\ \sigma = &\sqrt{\ln\left(1 + \frac{36}{25} v^2\right)} \, ,\end{aligned} \tag{4.2}$$

where μ and v are drived from

$$\begin{aligned}\hat{Z} =& \frac{\ln(\sigma_t) - \ln(\sigma_0)}{\nu\sqrt{t}} \\ m_k(\hat{Z}) =& \frac{\Phi(\hat{Z} + k\nu\sqrt{t}) - \Phi(\hat{Z} - k\nu\sqrt{t})}{2k\nu\sqrt{t} \, \phi(\sqrt{\hat{Z}^2 + k^2\nu^2 t})} \\ c(\hat{Z}) =& \tfrac{1}{2}\left(\frac{\sigma_t}{\sigma_0} + \frac{\sigma_0}{\sigma_t}\right) \\ \mu =& \frac{\sigma_t}{\sigma_0} \, m_1(\hat{Z}) \\ v =& \left(\frac{\sigma_t}{\sigma_0}\right) \frac{\sqrt{m_2(\hat{Z}) - c(\hat{Z}) \, m_1(\hat{Z})}}{\mu \, \nu\sqrt{t}} - \mu \, ,\end{aligned} \tag{4.3}$$

where $\Phi(.)$ and $\phi(.)$ are standard normal CDF and PDF, respectively. Significant testing showed that while the approximation approach of (4.2) works reasonably well for t less than three years, it tends to diverge for larger time steps.

A more robust approach for the sampling of I_0^t is the exact approach of Cai *et al*[4], which is based on Laplace transform technique. In their paper they showed that the Laplace transform $L(\theta)$ of the CDF of $1/I_0^t$ is given by

$$\begin{aligned} L(\theta) &= \mathcal{L}(\theta; F_{1/I_0^t}(u)) = \int_0^\infty F_{1/I_0^t}(u)\, \exp(-\theta\, u)\, du \\ L(\theta) &= \frac{\sigma_0^2 t}{\theta}\, \exp\left\{ \frac{[\ln(\sigma_t/\sigma_0)]^2 - [\psi(\theta\nu^2/\sigma_0^2)]^2}{2\nu^2\, t} \right\} \\ \psi(x) &= \operatorname{arcosh}\left(\frac{x\sigma_0}{\sigma_t} + \tfrac{1}{2}(\frac{\sigma_t}{\sigma_0} + \frac{\sigma_0}{\sigma_t}) \right)\ . \end{aligned} \tag{4.4}$$

Several powerful numerical algorithms have been proposed in the literature for the inversion of Laplace transforms. We present here the Euler algorithm proposed by Abate and Whitt (2006)[1] for the inversion of (4.4), which reads

$$\begin{aligned} F_{1/I_0^t}(t) &= \frac{10^{\frac{M}{3}}}{t} \sum_{k=0}^{2M} (-1)^k \eta_k\, \Re\left(L(\frac{\beta_k}{t}) \right) \\ \beta_k &= \frac{1}{3}\, M \ln(10)\ +\ \pi i k \\ \eta_0 &= \frac{1}{2}, \qquad \eta_k = 1,\ \ 1 \le k \le M \qquad \eta_{2M} =\ \ 2^{-M} \\ \eta_{2M-k} &= \eta_{2M-k+1}\ +\ 2^{-M}\binom{M}{k},\ \ 0 < k < M\ , \end{aligned} \tag{4.5}$$

where $i = \sqrt{-1}$, $\Re(.)$ is the real part of a complex number and M is chosen according to the desired level of accuracy. The number of correct significant digits is greater than $0.6M$, for well-behaved transforms. Notice that the CDF of I_0^t is equal to $1 - F_{1/I_0^t}$, since $I_0^t > 0$, and hence, the relationship between the two is a bijection and

$$\operatorname{sign}(\frac{d(1/I_0^t)}{dt}) = -\operatorname{sign}(\frac{dI_0^t}{dt})\ .$$

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5 Sampling the CEV Process

Since the SABR model is a stochastic volatility generalization of the Constant Elasticity of Variance (CEV) model, sampling CEV variates is usually a needed step in the SABR simulation.

Definition 1 (CEV) *A CEV distribution which is denoted by*

$$F_t \sim \mathcal{CEV}_\beta(F_0, \sigma_0^2 t)\,, \tag{5.1}$$

is the distribution of F_t, which follows the stochastic process

$$dF_t = \sigma_0 F_t^\beta dW_t \quad \sigma \geq 0,\ F_0 \geq 0,\ 0 \leq \beta \leq 1\,, \tag{5.2}$$

where dW_t is a standard Wiener process.

Islah, 2009[9], showed that a SABR model with $\rho = 0$ is reducible to a CEV model by replacing $\sigma_0^2 t$ by I_0^t, and that the CEV CDF follows the following probability law

$$\begin{aligned}
\mathbb{P}(F_t = 0 \big| F_0, \sigma_0, \sigma_t, I_0^t) &= 1 \;-\; Q_{\chi^2}\left(A_0; \frac{1}{1-\beta}\right) \\
\mathbb{P}(F_t \leq u \big| F_0, \sigma_0, \sigma_t, I_0^t) &= 1 \;-\; Q_{\chi'^2}\left(A_0; \frac{1}{1-\beta}; C_0(u)\right) \quad u > 0\,,
\end{aligned} \tag{5.3}$$

where

$$A_0 = \frac{1}{I_0^t}\left(\frac{F_0^{1-\beta}}{1-\beta}\right)^2 \text{ and}$$

$$C_0(u) = \frac{1}{I_0^t}\left(\frac{u^{1-\beta}}{1-\beta}\right)^2\,,$$

and where $\chi^2(\delta)$ is the chi-squared distribution with δ degrees of freedom, and $\chi\prime^2(\delta, \gamma)$ is the non-central chi-squared distribution with δ degrees of freedom and γ non-centrality parameter. The sampling of F_t requires the numerical inversion of Equation (5.3), a process fraught with difficulty due to wide range of the parameters involved which lead to high degree of numerical instability.

A novel approach proposed by Choi *et al*,[2] based on the work of Makarov and Glew[12] and Kang[10] allows for the direct sampling of a CEV variate through the drawing of two gamma samples and one Poisson sample. Using their approach, we define

$$\begin{aligned}
z_t &:= z(F_t) \\
z(y) &:= \frac{u^{2(1-\beta)}}{(1-\beta)^2 \sigma_0^2 t I_0^t}\,.
\end{aligned}$$

A sample of z_t can be drawn according to

$$\begin{aligned}
&Z_t \sim 2\mathcal{G}(N+1) \\
&N \sim \mathcal{POIS}(\frac{z_0}{2} - X) \\
&X \sim \mathcal{G}\left(\frac{1}{2(1-\beta)}\right) \\
&F_t = \begin{cases} 0 & X \geq 0.5\, z_0 \\ \left(\sigma_0(1-\beta)\sqrt{t\, Z_t\, I_0^t}\right)^{\frac{1}{1-\beta}} & X < 0.5\, z_0\,, \end{cases}
\end{aligned} \tag{5.4}$$

where $\mathcal{G}(\alpha)$ is the gamma distribution with shape parameter α and unit scale parameter, and $\mathcal{POIS}(\lambda)$ is the Poisson distribution with intensity λ.

6 Approximating the SABR Process

As already mentioned in Section 2, the SABR process can be reduced to a CEV process, with $\sigma_0\sqrt{t\,I_0^t}$ playing the rôle of σ_t, whenever $\rho = 0$. As of the time of this writing, however, there is no known analytic expression of the general case of SABR dynamics with $\rho \neq 0$. In this section we present an approximation, based on an operator splitting technique, as suggested by Choi em et al[2]. According to this approach we rewrite Equation (1.1) as

$$\frac{dF_t}{F_t^\beta} = \sigma_t\,(\rho\,dZ_t + \sqrt{1-\rho^2}\,dZ_t^\perp) \quad \text{and} \quad \frac{d\sigma_t}{\sigma_t} = \nu\,dZ_t\,, \tag{6.1}$$

where dZ_t and $dZ_t^\perp$ are uncorrelated Wiener increments. Applying the operator splitting technique Equation (6.1) becomes

$$\frac{dF_t^{(1)}}{[F_t^{(1)}]^\beta} = \rho\,\sigma_t\,dZ_t \quad \text{and} \quad \frac{d\sigma_t}{\sigma_t} = \nu\,dZ_t \tag{6.2a}$$

$$\frac{dF_t^{(2)}}{[F_t^{(2)}]^\beta} = \sqrt{1-\rho^2}\,\sigma_t\,dZ_t^\perp \quad \text{and} \quad \frac{d\sigma_t}{\sigma_t} = \nu\,dZ_t\,. \tag{6.2b}$$

The operator splitting technique approximates the solution F_t of (6.1), with initial condition F_0, by $F_t \approx F_t^{(2)}\left(F_t^{(1)}(F_0)\right)$, where $F_t^{(1)}(F_0)$ is the solution $F_0^{(1)}$ of (6.2a), with initial condition $F_0^{(1)} = F_0$, and $F_t^{(2)}(F_t^{(1)}(F_0))$ is the solution $F_t^{(2)}$ of Equation (6.2b), with initial condition $F_0^{(2)} = F_t^{(1)}(F_0)$.

According to Section 2, Equation (6.2b) which obeys the SABR dynamics with zero correlation, is essentially a CEV process whose solution is

$$F_t^{(2)}|\sigma_t, I_0^t \sim \mathcal{CEV}_\beta\left(F_0^{(2)}, (1-\rho^2)\sigma_0^2 t I_0^t\right)\,. \tag{6.3}$$

This leaves us with Equation (6.2a), which has no known analytic solution. Choi em et al[2] suggested a frozen coefficient approximation in which they approximated $[F_t^{(1)}]^{(1-\beta)}$ by $F_0^{(1-\beta)}$, accordingly,

$$\begin{aligned}\frac{dF_t^{(1)}}{F_t^{(1)}} &= \frac{\rho\sigma_t}{[F_t^{(1)}]^{(1-\beta)}}\,dZ_t \\ &\approx \frac{\rho\sigma_t}{F_0^{(1-\beta)}}\,dZ_t\,.\end{aligned} \tag{6.4}$$

The solution $\tilde{F}_t^{(1)}$ of Equation (6.4) is

$$\tilde{F}_t^{(1)}(F_0) = F_0\,\exp\left(\frac{\rho}{\nu F_0^{(1-\beta)}}(\sigma_t - \sigma_0) - \frac{\rho^2\sigma_0^2 t I_0^t}{2F_0^{2(1-\beta)}}\right)\,. \tag{6.5}$$

Finally, conditional on σ_t and I_0^t, the CEV approximation of the general SABR process results in a terminal forward price F_t according to

$$F_t|\,\sigma_t, I_0^t \sim \mathcal{CEV}\left(\tilde{F}_t^{(1)},\,(1-\rho^2)\sigma_0^2\,t\,I_0^t\right) \tag{6.6}$$

As is the case with almost all approximation approaches of this nature, the accuracy of the final result increases in inverse proportion to the time step t.

7 Implied Volatility

Given an option pricing model, the *implied volatility* is a value, reflective of the variance of the stochastic process, that, when plugged into the model, gives the observed option price. Derivatives traders prefer implied volatility as a price quotation mechanism. This is due to the fact that it can provide stable option quotation that does not need to change with every movement of the underlying. Moreover, implied volatility reveals the nature of option prices as a function of option strike—generally referred to as the volatility smile.

Let $B(\sigma; u, k, \tau)$ be the price of an option of strike k, on an underlying of forward price u and price volatility σ, and which expires at time τ. Given a specific tuple (v, u_0, k, τ), where v the price of the option at time $t = 0$, the implied volatility of the option is

$$\sigma^{imp} = V(v; u_0, k, \tau)\,, \tag{7.1a}$$

that satisfies

$$B(V(v; u_0, k, \tau); u, k, \tau) = v\,, \tag{7.1b}$$

i.e. , $V(.) = B(.)^{-1}$.

When the dynamics of the underlying are believed to be governed by a geometric Brownian motion, the pricing model is log-normal and the implied volatility is often referred to as the Black or Black-Scholes volatility, σ^{BS}. For instance, in the case of a call option, σ^{BS} can be obtained from

$$B(\sigma; u_0, k, \tau) = u_0\,N(d_1)\; - \;k\,N(d2) \qquad \text{with} \qquad d_{1,2} = \frac{\log(u_0/k)}{\sigma^{BS}\sqrt{\tau}} \pm \frac{\sigma^{BS}\sqrt{\tau}}{2}\,, \tag{7.2}$$

where $N(.)$ is the CDF of the standard Gaussian distribution.

Similarly, when the dynamics are Gaussian in nature, the pricing model is Bachelier and the implied volatility is known as the normal volatility, σ^N, which in the case of a call option should satisfy

$$B(\sigma; u_0, k, \tau) = (u_0 - k)\,N(d_N)\; + \;\sigma^N\sqrt{\tau}\,n(d_N) \qquad \text{with} \qquad d_n = \frac{u_0\; - \;k}{\sigma^N\sqrt{\tau}}\,, \tag{7.3}$$

where $n(.)$ is the PDF of the standard Gaussian distribution.

More relevant to the SABR model is the notion of the CEV implied volatility, σ^{CEV}, which is the volatility associated with the constant elasticity of variance model. In the case of a call option, σ^{CEV} obeys (see [5])

$$
\begin{aligned}
B(\sigma; u_0, k, \tau) = \bar{F}_{\chi^2}\left(\frac{k^{2\beta_*}}{[\sigma^{CEV}\beta_*]^2\,\tau}; 2+\frac{1}{\beta_*}, \frac{1}{[\sigma^{CEV}\beta_*]^2\,\tau}\right) - \\
k\,F_{\chi^2}\left(\frac{1}{[\sigma^{CEV}\beta_*]^2\,\tau}; \frac{1}{\beta_*}, \frac{k^{2\beta_*}}{[\sigma^{CEV}\beta_*]^2\,\tau}\right),
\end{aligned}
\tag{7.4}
$$

where $\beta_* = 1-\beta$, $\bar{F}_{\chi^2}(.) = 1 - F_{\chi^2}(.)$ and $F_{\chi^2}(x; r; x_0)$ is CDF of the non-central chi-squared distribution of r degrees of freedom and x_0 non-centrality parameter.

It goes without saying that any implied volatility can be converted to any other implied volatility. This is done by plugging the given implied volatility into the corresponding option pricing model. Once the option price is obtained, it can be used to obtain the desired implied volatility by, usually numerically, solving the relevant inverse pricing model $B^{-1}(.)$.

8 Asymptotic Representations

Since deriving closed form analytic formulas for derivatives pricing is, most of the time, impossible or impractical, a significant amount of research effort has been directed towards finding effective semi-analytic approximations. These are usually formulas for expressing implied volatility in terms of model parameters. In this section, we focus on semi-analytic representations of the SABR model. All formulas presented in this section will use the following normalized quantities,

$$
\begin{aligned}
f_t = &\frac{F_t}{F_0} \quad (f_0 = 1) \quad k = \frac{K}{F_0} \\
\hat{\sigma}_t = &\frac{\sigma_t}{\sigma_0} \quad (\hat{\sigma}_0 = 1) \quad \alpha = \frac{\sigma_0}{F_0^{\beta_*}},
\end{aligned}
\tag{8.1}
$$

where K is the option strike price. We will denote option expiry by T.

8.1 Hagan *et al*'s (HKLW) Formula

The Hagan approximation (2002)[6] is one of the most widely used semi-analytic approximations of the SABR model. It is very effective near the money and for small times to maturity. Hagan *et all* presented $O(T)$ approximations for both the normal and the Black-Scholes volatilities. Their normal approximation is more accurate. In a later paper, Hagan *el al* (2014)[7], derived a modified more accurate version of their earlier approximation. In the modified Hagan model, the normal implied volatility of the SABR model is given by the following chain of definitions:

$$
\begin{aligned}
\sigma^N =&\alpha \frac{q_N}{q} H(z)\,(1\,+\,h_N\,T), \qquad z=\frac{\nu}{\alpha}\,q \\
q =& \int_1^k k^{-\beta}\,dk = \begin{cases} \frac{k^{\beta_*}-1}{\beta_*} & \text{if} \quad 0\le\beta<1 \\ \log k & \text{if} \quad \beta=1 \end{cases} \\
q_N =&k-1 \qquad (z_N=\frac{\nu}{\alpha}\,q_N) \\
H(z) =& \frac{z}{x(z)}, \quad x(z)=\,\log\left(\frac{V(z)+z+\rho}{1+\rho}\right), \quad V(x)=\sqrt{1+2\rho z+z^2} \\
h_N =& \log\left(k^{\beta/2}\frac{q}{q_N}\right)\frac{\alpha^2}{q^2}\,+\,\frac{\rho}{4}\frac{k^\beta-1}{k-1}\alpha\nu\,+\,\frac{2-3\rho^2}{24}\nu^2\,.
\end{aligned}
\tag{8.2}
$$

There are three special cases concerning Equation (8.2). First, when $z=0$, $H(z)$ converges to 1. Secondly, when $\rho=0$, $H(z)$ simplifies to $H(z)=z/arcsinh(z)$. Finally, the third case is the case of $\rho=-1$. In such a case, the value of $H(z)$ becomes

$$
H(z)=\begin{cases} 0 & \text{if} \quad z=0 \\ \infty & \text{otherwise}\,. \end{cases} \tag{8.3}
$$

8.2 Choi and Wu's CEV Formula

In 2021 Choi and Wu[5] presented a powerful formula for the approximation of σ^{CEV}, based on Paulot's (2015) work[14]. This formula is better at handling out of the money options, large initial volatility and large vol-of-vol.

9 Control Variable

It is not uncommon in a risk management scenario for the set $I_1 \times I_2$ to have a size in the hundreds. Estimating portfolio risk requires the simulation of the volatility surface over hundreds of forward horizons. It is not hard to see how the computational complexity grows very rapidly. As a result, each simulation of the volatility surface has to be limited to a number of paths that is far less than that which needed for accurate simulation of an option price. To alleviate that deficiency, we resort to a technique know as the *control variable* technique as a means of variance reduction and, consequently, accuracy improvement.

Our control variable approach uses the prices of four control options struck at different strikes, for every node on the implied volatility grid and for every simulation horizon, to adjust the distribution of the forward price F_h of the simulated sample paths. The philosophy behind this stems from the fact that the true forward price distribution can be ascertained from the option prices of options struck at different strikes. Given a forward price $F_0(t)$ and an implied volatility set $S_0(t,\tau)$, one can accurately price options of various strikes that share the tenor t and the expiry h. This is usually done using a semi-analytic technique, such as those discussed in Section 6, additional Monte Carlo simulation or a partial differential equation (PDE) solver such as the finite difference method (FDM). Once the option prices are obtained, we compare their values with those predictable by our simulation. The discrepancy, is then, corrected by proportionately adjusting the values of F_h of the simulation paths.

10 Bayesian Volatility

Assuming a standard filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, let the processes $F_t(F_0, \sigma_t, I_t),\ \sigma_t(\sigma_0)$ and $I_t(\sigma_t)$ be $\mathcal{F}_t$-measurable according to the SABR dynamics, stated in the preceding sections. In this section, we are concerned with finding the best estimate for σ_t, where F_t has been generated from an exogenous process. Using Bayesian statistics, we can write

$$P(\sigma_t = s|F_t = f) = \frac{P(F_t = x|I_t = a, \sigma_t = s)P(I_t = a|\sigma_t = s)P(\sigma_t = s)}{P(F_t = f)}\,. \tag{10.1}$$

Or more precisely,

$$P(\sigma_t = s|F_t = f) = \frac{P(F_t = f|I_t \in da, \sigma_t \in ds)P(I_t = a|\sigma_t \in ds)P(\sigma_t = s)}{P(F_t = f)} \tag{10.2a}$$

$$P(F_t = f) = \int_0^\infty\int_0^\infty P(F_t = f|I_t \in da, \sigma_t \in ds)\, P(I_t = a|\sigma_t \in ds)\, P(\sigma_t = s)\, da\, ds\,. \tag{10.2b}$$

Since we have already shown, in the early part of this paper, how to compute the distributions $P(F_t = f|I_t \in da, \sigma_t \in ds),\ P(I_t = a|\sigma_t \in ds)$ and $P(\sigma_t = s)$, we can now, at least numerically, compute the distribution $P(\sigma_t = s|F_t = f)$ and its CDF $Q_\sigma(s|F_t = f)$ defined as,

$$\begin{aligned} Q_\sigma(s|F_t = f) &= P(\sigma_t \leq s|F_t = f) \\ &= \int_0^s P(\sigma_t = s|F_t = f)\, ds\,. \end{aligned} \tag{10.3}$$

To sample the distribution $Q_\sigma(.)$, we need to compute $Q_\sigma^{-1}(U)$, where $U \sim Uniform(0,1)$ and Q_σ^{-1} is the inverse of $Q_\sigma(.)$

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